

Research Portfolio

Cecilia Andrea Callejas Pastor
세실리아 칼레하스

PhD. In Biomedical Engineering
Research Professor
Chungnam National University Hospital

✉ cecilia.callejas.pastor@gmail.com

☎ +82 10-4302-9697

📍 South Korea



CECILIA ANDREA CALLEJAS PASTOR

세쉴리아 칼레하스

Biomedical engineer specializing in AI-driven algorithm and software medical technologies. Passionate about applying cutting-edge technology to healthcare and wellness devices. Committed to continuous learning and contributing to the advancement of healthcare technology worldwide!

EXPERIENCE

Present	Research Professor at Chungnam National University Hospital
2025	Postdoctoral Researcher at Seoul National University & Hospital
2024	Postdoctoral Researcher at Chungnam National University
2024	Senior Researcher Intern at MedInTech Inc.

EDUCATION

2024	PhD in Biomedical Engineering, Chungnam National University
2021	MSc in Biomedical Engineering, Chungnam National University
2016	BSc in Biomedical Engineering, Bolivian Catholic University

KEY SKILLS

Languages: Spanish (Native), English (TOEIC 970), Korean (TOPIK 5), Italian
Software: Python, MATLAB, Android Studio/Java, Flutter/Dart, Jira Cloud, REST API
Hardware: SolidWorks, OrCAD, PADS
Firmware: Segger Embedded Studio, STM32 CubeMX

PUBLICATIONS

Journal Articles: 7 (4 published, 3 under review) > 6 as first author
Conference Presentations: 9
Patents: 5 (4 completed, 1 under process)

KEY ACHIEVEMENTS

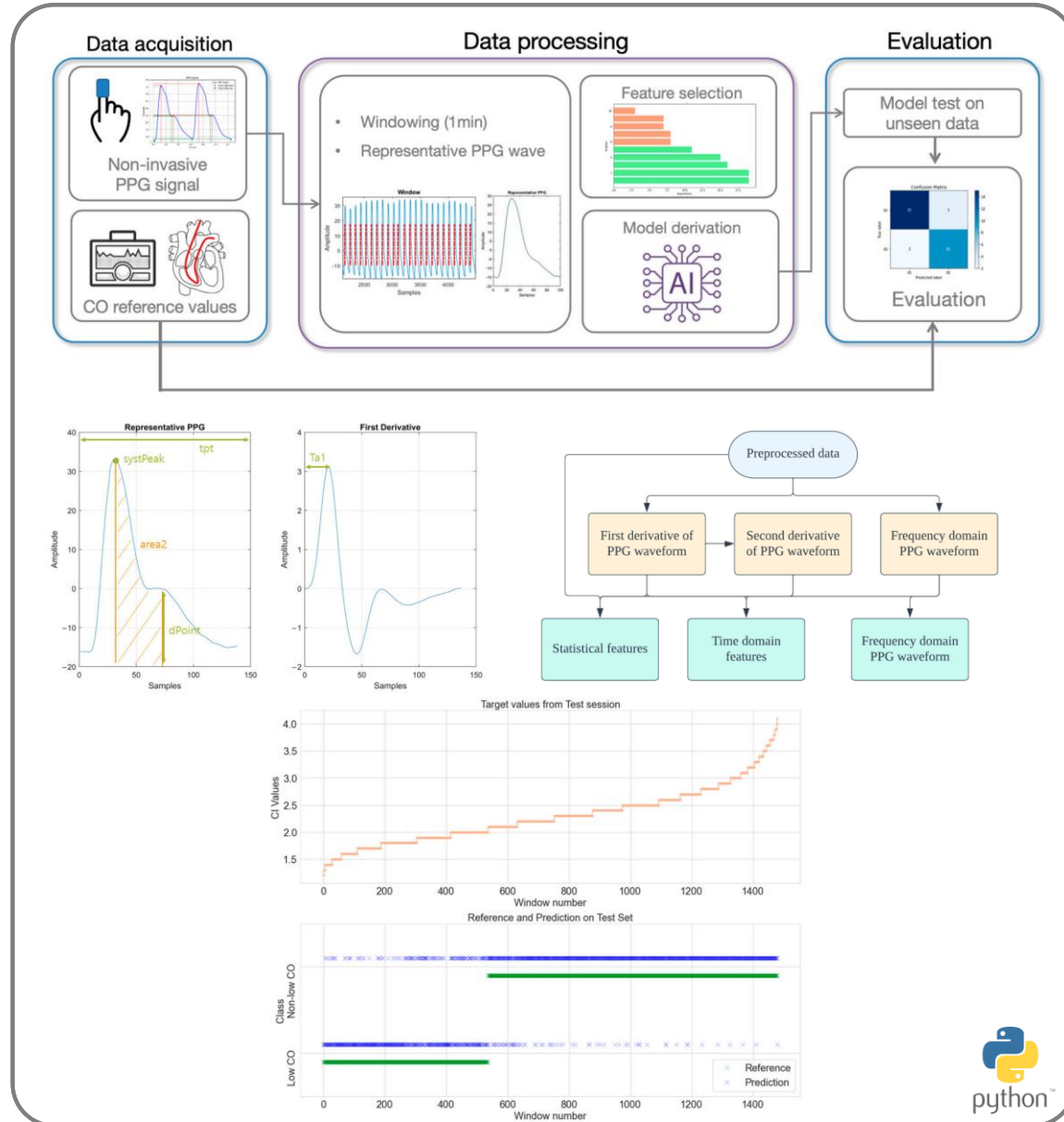
2025	Best Poster Award - Korean Balance Society
2024	Best Poster Award - Korean Innovative Medical Technology Society
2023	Best Oration Award - Korean Balance Society
2022	AI & BigData Scholar - Daewoong Foundation (2022-2024)
2021	Global Korean Scholarship Program Scholar - PhD Program (2021-2024)
2021	AI & BigData Hackathon by Daewoong Foundation - Third Place
2021	Best Research Award - Chungnam National University
2018	Global Korean Scholarship Program Scholar - Masters Program (2018-2024)

Contents

- Research in healthcare applications based on physiological signal analysis
- Research on AI-based diagnostic tools
- Research on healthcare wearable system

Research in healthcare applications based on physiological signal analysis

PPG-Based Cardiac Output Estimation

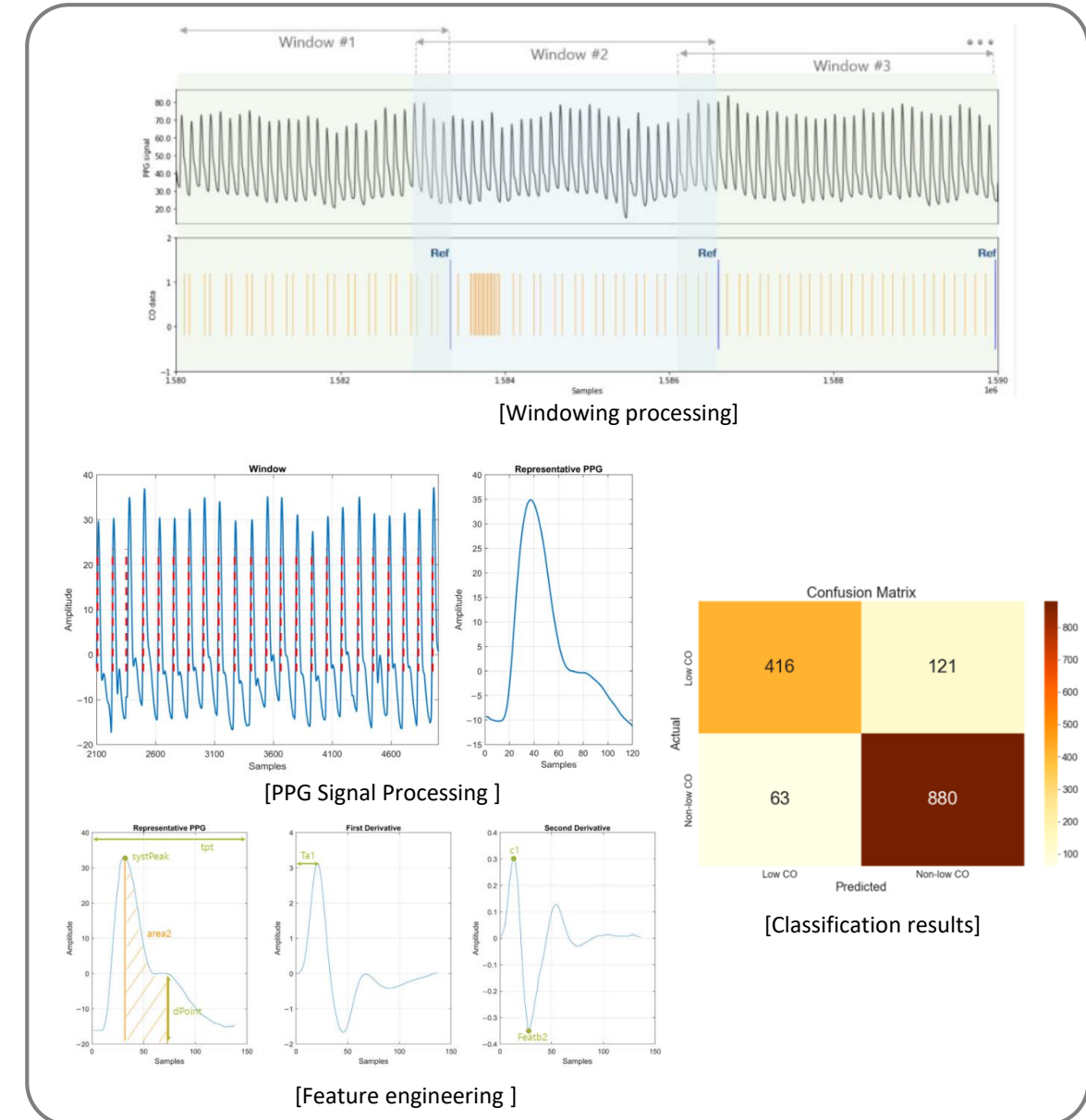


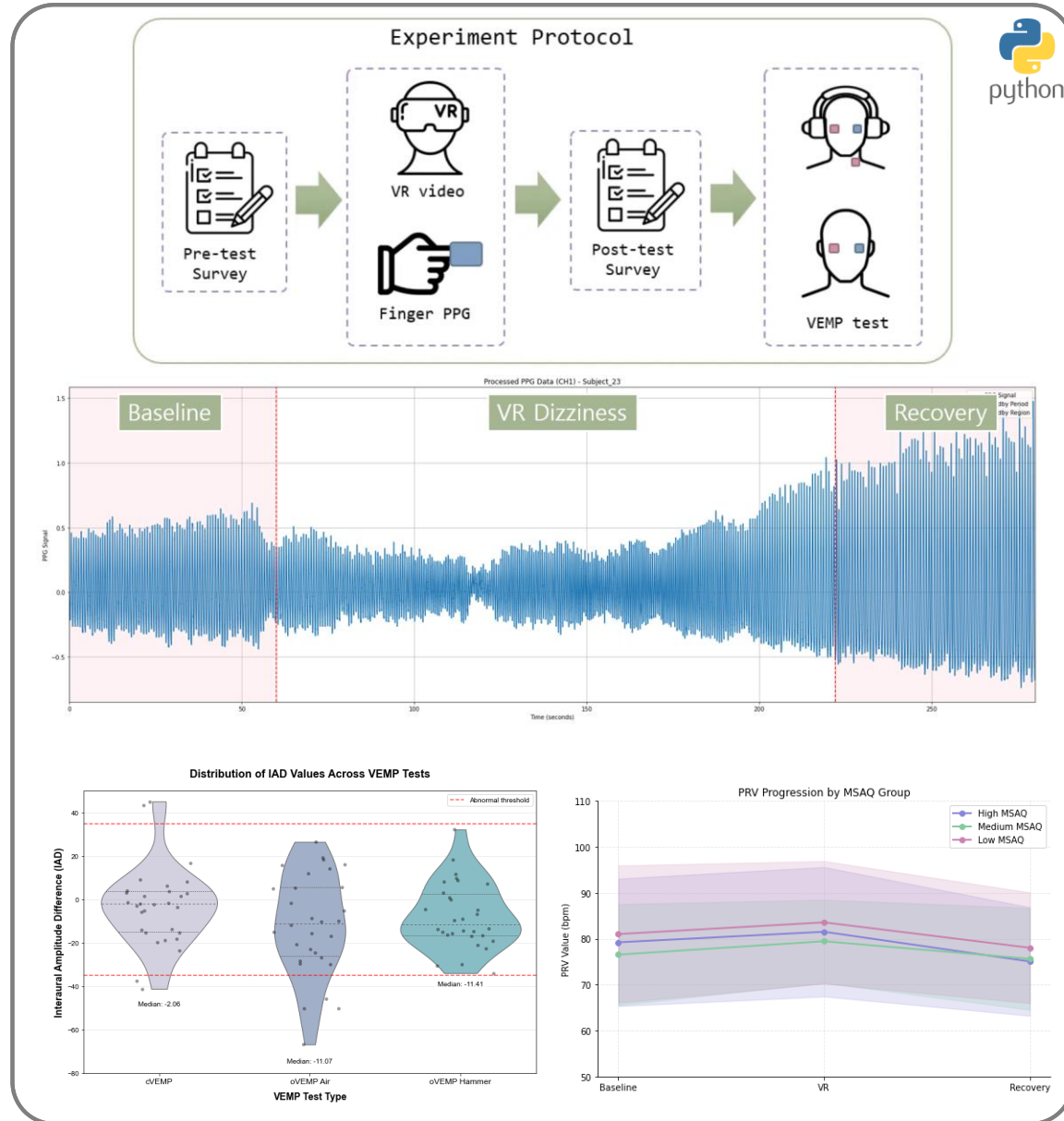
MACHINE LEARNING-BASED CARDIAC INDEX ESTIMATION USING PHOTOPLETHYSMOGRAPHY

- **Objective:** To develop a non-invasive algorithm to classify cardiac index (CI) using finger photoplethysmography (PPG) and machine learning as an alternative to invasive thermodilution methods.
- **Methods:** The system aimed to accurately identify low versus non-low cardiac output states in patients undergoing off-pump coronary artery bypass surgery
 - Developed non-invasive cardiac output algorithm for classification
 - Used Relief algorithm for optimal feature selection
- **Results:**
 - Implemented CatBoost model achieving 87.57% test accuracy
 - [Published in Journal of Clinical Medicine \(2024, SCIE Q1\)](#)
- **Skills developed:** Feature engineering, ML model selection and optimization, time-series signal processing, physiological data analytics

PPG-Based Cardiac Output Estimation

- **Reference Standard:** Used pulmonary artery catheter thermodilution as gold standard for comparison of 67 patients undergoing off-pump coronary artery bypass surgery
- **PPG Signal Processing:**
 - Collected finger PPG at 125 Hz from patients during cardiac surgery
 - Applied bandpass filtering (0.1-15 Hz) and detrending
 - Created representative waveforms by averaging cardiac cycles
- **Feature Engineering:**
 - Extracted 20 key features from PPG waveform and its derivatives
 - Analyzed systolic peaks, diastolic notches, and spectral components
 - Used Relief algorithm for optimal feature selection
- **Classification Framework:**
 - Categorized cardiac index into "Low CO" (≤ 2 L/min/m²) vs. "Non-low CO"
 - Implemented CatBoost model with 5-fold cross-validation
 - Validated against gold-standard thermodilution measurements



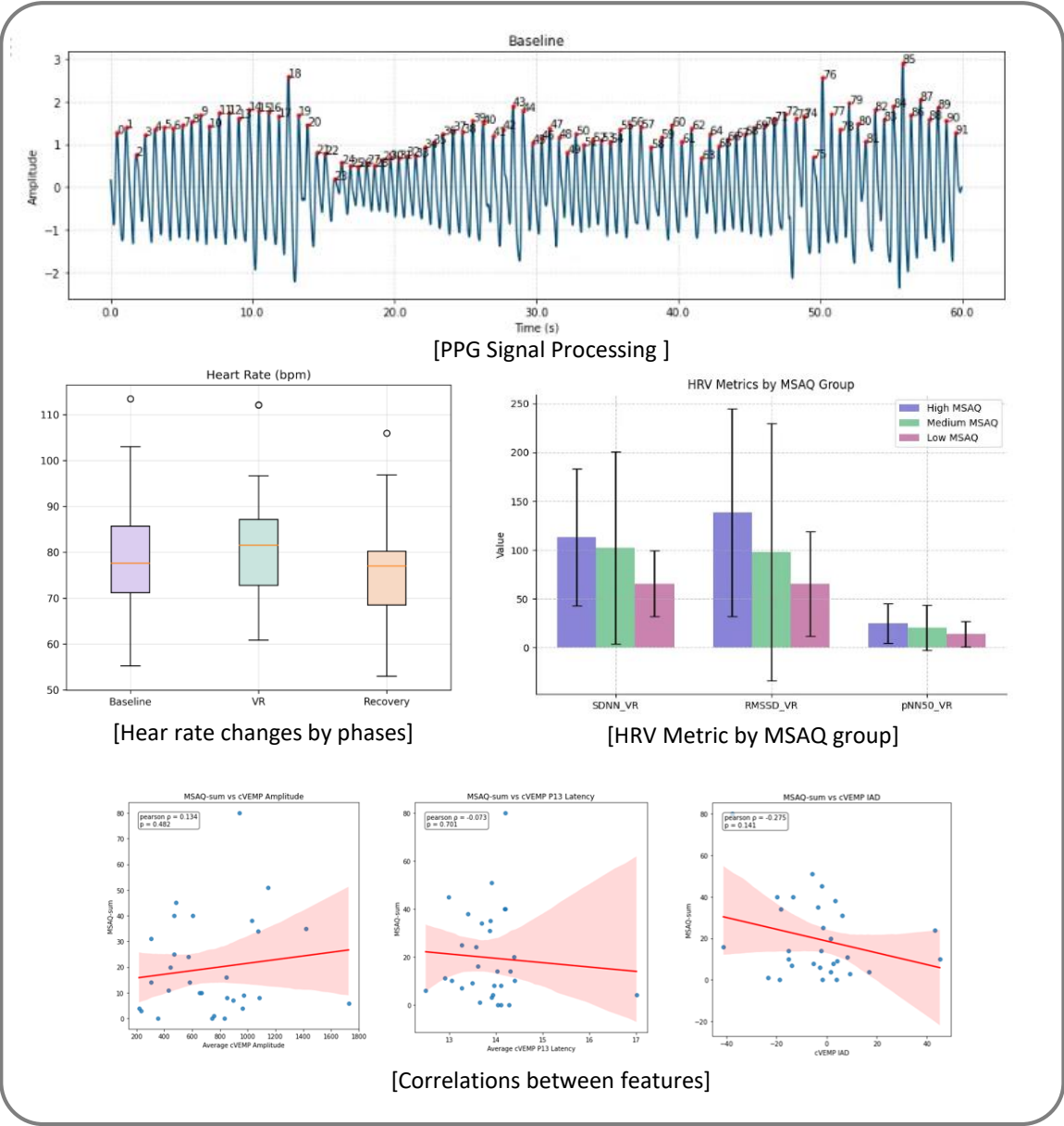


DEVELOPMENT OF A CYBER-SICKNESS FREE SYSTEM TO OVERCOME THE LIMITATION OF XR DIGITAL THERAPEUTICS

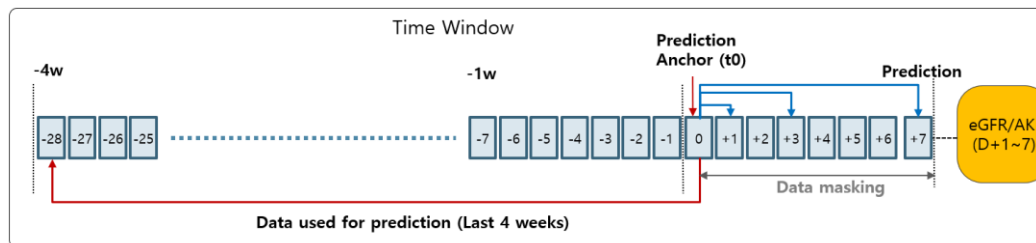
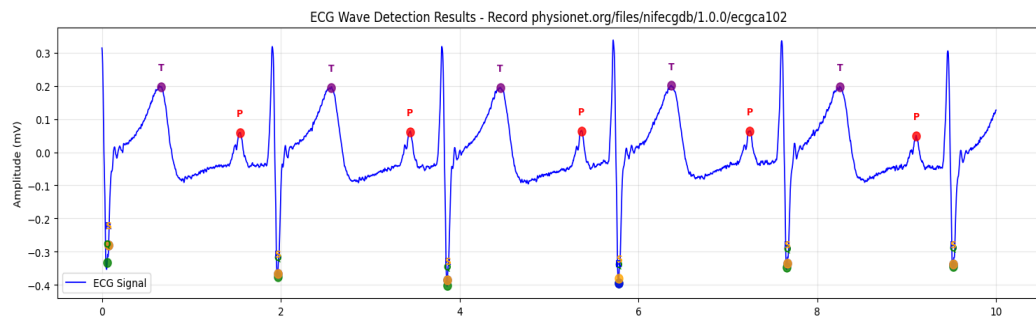
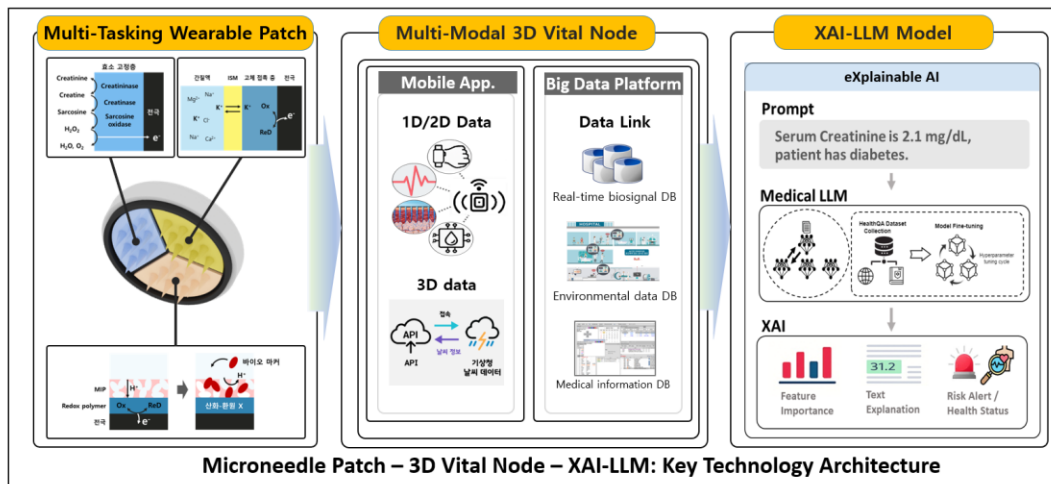
- **Objective:** To develop an objective measure of cyber-sickness in XR digital therapeutics by analyzing physiological responses during virtual reality exposure.
- **Methods:** Data analysis between non-invasive finger PPG-based monitoring and autonomic nervous system changes
 - Analyzed pulse rate variability (PRV) as potential objective indicator of cybersickness
 - Investigated correlation between PRV and VEMP
- **Results:**
 - Observed significant HR changes during VR (+3.67%) and recovery (-6.30%) phases
 - Found correlations between VR sickness severity and vestibular function ($r=0.551$, $p=0.002$)
 - Under review in [Journal of Clinical Neurology \(2025, SCIE Q2\)](#)
- **Skills developed:** Autonomic nervous system analysis, PPG signal processing, clinical correlation studies, VR-related biomarker research

Cybersickness Detection Research

- **Reference Standard:** Used validated questionnaires (MSAQ, MSS-F, MSS-S, VR-SSS) and vestibular evoked myogenic potentials (VEMP) to assess motion sickness susceptibility in 30 participants
- **PPG Signal Processing:**
 - Recorded finger PPG at 2000 Hz during baseline (60s), VR exposure (162s), and recovery (60s)
 - Extracted heart rate variability parameters across experimental phases
 - Segmented data to analyze autonomic nervous system response patterns
- **Feature Analysis:**
 - Measured heart rate variability metrics (HR, SDNN, RMSSD) across all phases
 - Performed statistical comparisons between phases using paired t-tests
 - Grouped participants by motion sickness susceptibility (high, medium, low)
- **Correlation Framework:**
 - Analyzed relationships between HRV parameters and subjective VR sickness reports
 - Identified correlations between vestibular function and motion sickness susceptibility
 - Evaluated potential biomarkers through statistical correlation analysis ($p < 0.05$)



ECG analysis for LLM-based detection Research



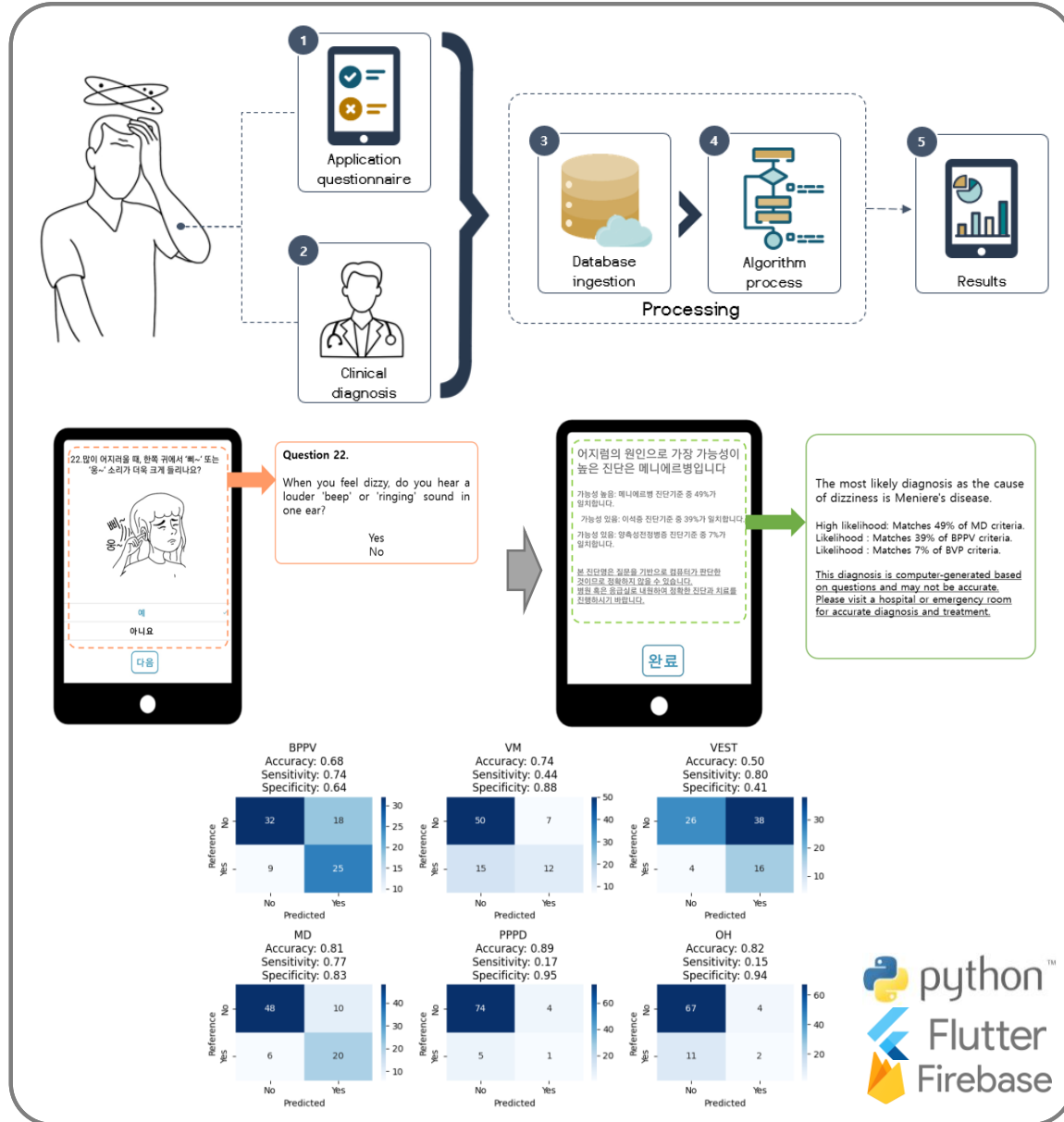
XAI-LLM-BASED MANAGEMENT OF CHRONIC KIDNEY DISEASE AND DETECTION OF ACUTE EXACERBATION

- **Objective:** To develop a explainable AI system for chronic kidney disease prevention and management, shifting from treatment-centered to patient-centered care in daily life.
- **Methods:** MIMIC-IV ECG Analysis: Extracted T-wave morphology features for hyperkalemia detection integrated with lab events, medication data, and I/O events
 - Healthcare wearable integration for continuous physiological data collection
 - Multi-modal liquid biopsy patch development for non-invasive biomarker monitoring
 - XAI-LLM Algorithm: Developed predictive models using 1-4 weeks prior patient data for eGFR trajectory and AKI episode forecasting
 - Data link framework connecting patch sensors, wearables, and clinical parameters
- **Results:**
 - Integrated ECG-based hyperkalemia detection with multi-parameter clinical features
 - Implemented LLM-based prediction system for future eGFR and acute kidney injury episodes
- **Skills developed:** Multi-modal biosignal processing (ECG, PPG), time-series clinical data analysis, explainable AI/LLM model development, healthcare system integration.



Research on AI-based diagnostic tools

AI-Based Vestibular Disorder Diagnostic System

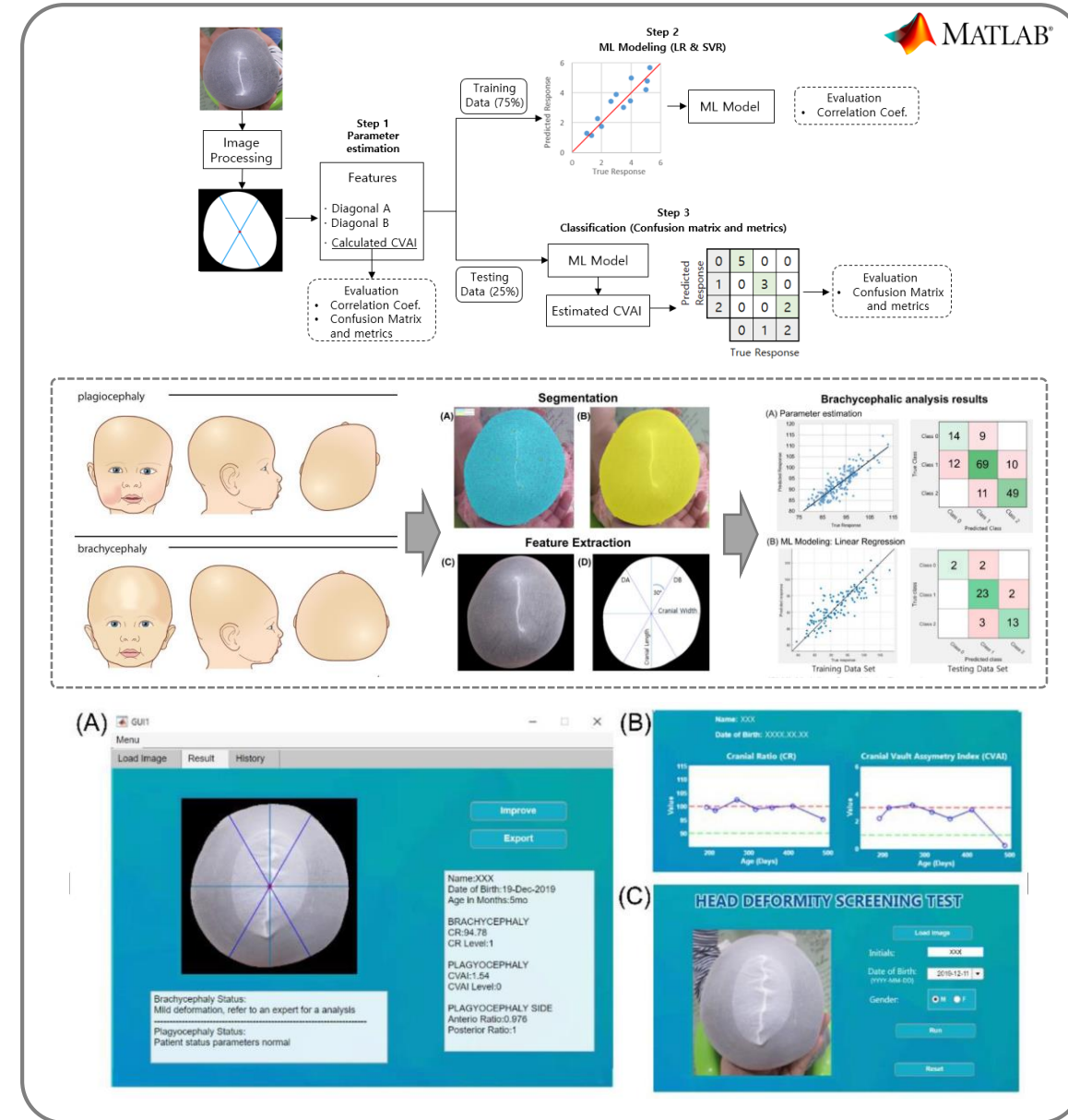


CLINICAL DECISION SUPPORT FOR VESTIBULAR DIAGNOSIS

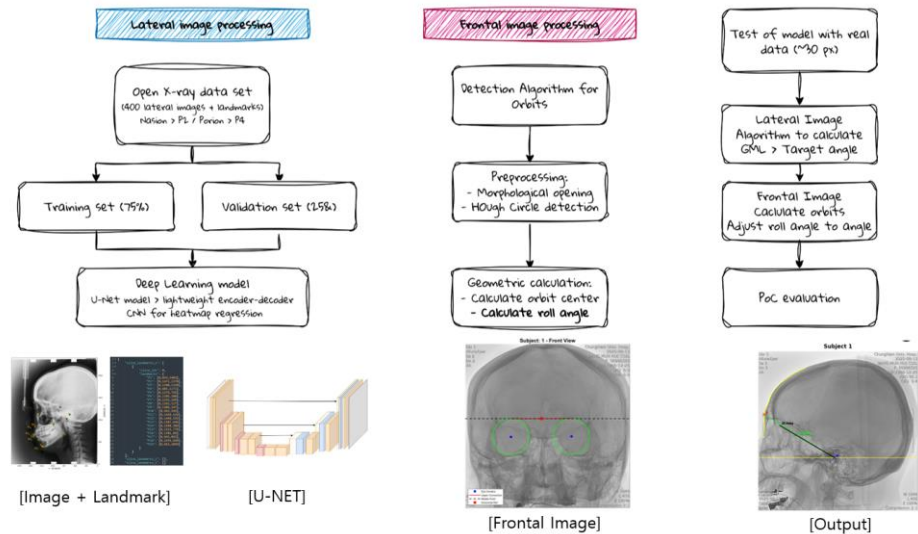
- Objective:** To develop a mobile application screening tool for six common vestibular disorders using rule-based & AI algorithms.
- Methods:** Developed Flutter-based mobile questionnaire system in collaboration with ENT specialists
 - Implemented a 47-item questionnaire to analyze symptoms
 - Created AI models analyzing 3,679 patient records
- Results:**
 - Achieved 82.1% diagnostic accuracy with rule-based algorithm (107 participants)
 - Improved to 86% classification accuracy with AI models
 - Patent application completed (KR10-2023-0055247)
 - ① Under review in Scientific Reports (2025, SCIE Q1)
 - ② Published in NPJ Digital Medicine (2025, SCIE Q1)
- Skills developed:** Android mobile application development, Flutter development, healthcare UI/UX design, clinical workflow implementation, diagnostic algorithm development, cross-platform mobile development, ML algorithms optimization

TWO-DIMENSIONIONAL IMAGE-BASED SCREENING FOR POSITIONAL CRANIAL DEFORMITIES

- **Objective:** To develop an innovative, cost-effective screening method for infant cranial deformities using 2D vertex images and machine learning algorithms.
- **Methods:**
 - Created MATLAB UI with automated image processing pipeline
 - Implemented image processing and contour detection
 - Extracted accurate Cranial Ratio and Cranial Vault Asymmetry Index measurements
- **Results:**
 - Achieved strong correlation with gold-standard 3D scanner (0.85-0.89)
 - Published in Diagnostics (2020, SCIE Q1)
 - Patent registered (KR2377364) , (KR2377363)
- **Skills developed:** MATLAB programming, medical image processing, feature extraction algorithms, user interface design and statistical analysis

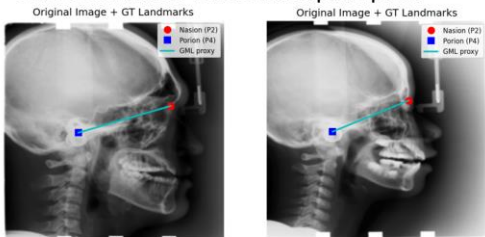


AI-Based Vestibular Disorder Diagnostic System

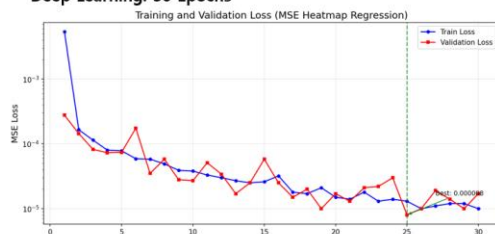


Training process:

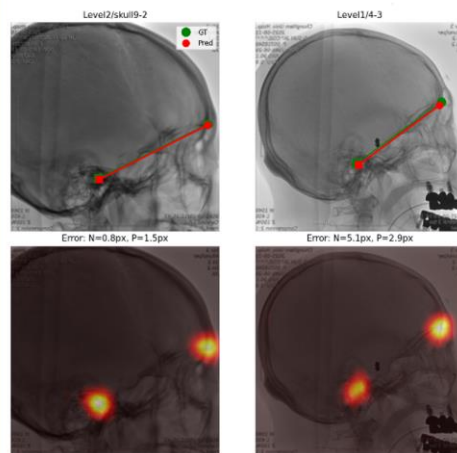
U-net + CNN decoder-encoder that outputs 2 points



Deep Learning: 30 Epochs



Test on our own results



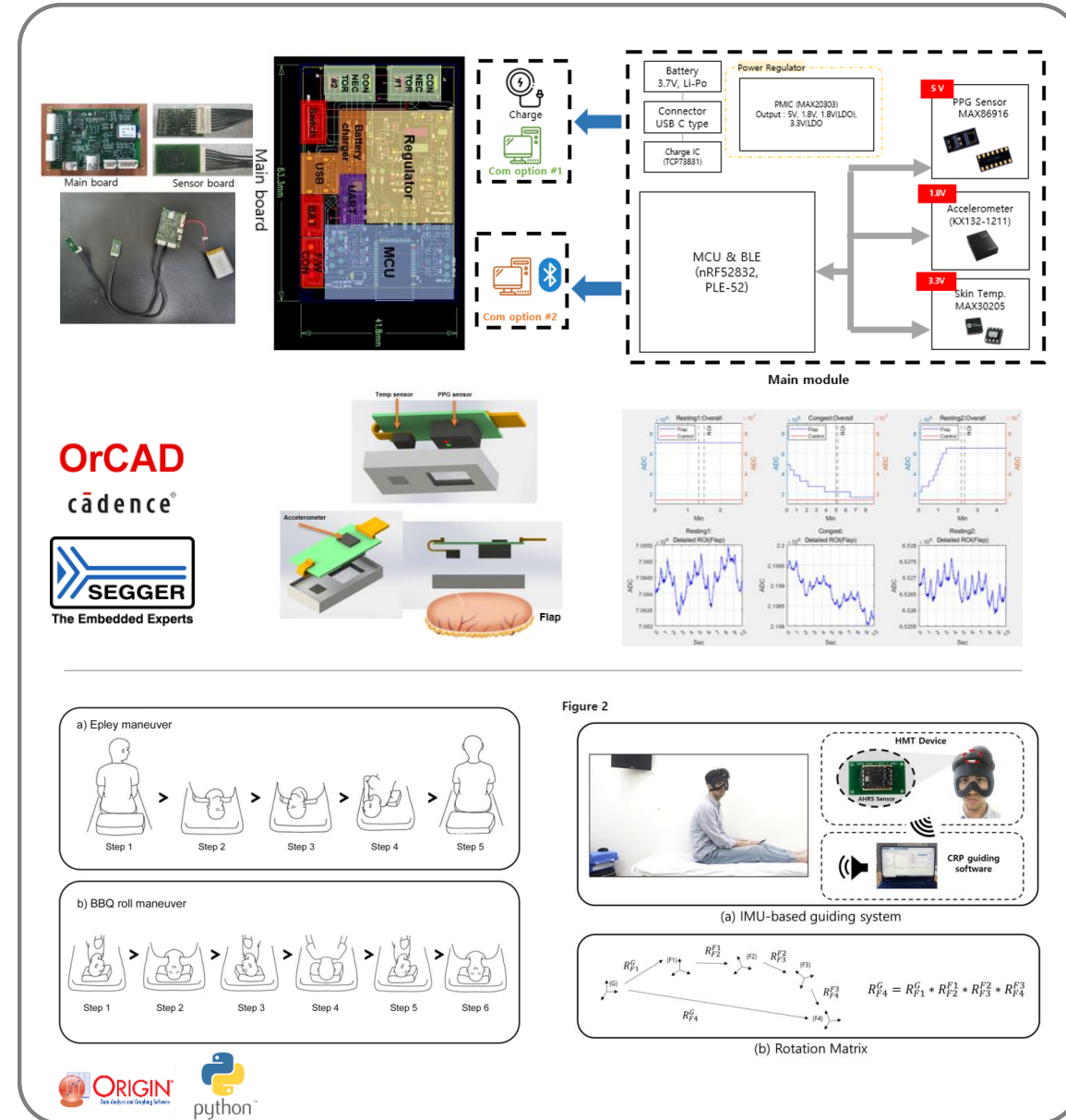
AI-BASED AUTOMATED IMAGING ANGLE SETTING SYSTEM FOR CEREBRAL ANGIOGRAPHY

- **Objective:** To develop an automated, objective imaging angle determination system for cerebral angiography
- **Methods:** Developed Flutter-based mobile questionnaire system in collaboration with ENT specialists
 - Image preprocessing: CLAHE contrast enhancement, Gaussian blur, morphological operations for anatomical landmark detection
 - Deep learning U-Net encoder-CNN decoder trained on 400 annotated images with spatial augmentation
 - Angle calculation: Geometric computation from orbit-meatal point baselines and coordinate transformation
- **Results:**
 - Clinical images and intentionally shifted/rotated images for robustness testing
 - Patent application completed (KR10-2026-0021401)
- **Skills developed:** Medical image preprocessing, anatomical landmark detection, U-Net architecture, encoder-decoder networks

Research on healthcare wearable system

DEVELOPMENT OF ADVANCED MEDICAL SENSING SYSTEMS FOR SURGICAL MONITORING AND VESTIBULAR TREATMENT

- **Objective:** To develop advanced medical devices for healthcare applications, including a smart flap monitoring system using AIoT and an IMU sensor-based guiding system for BPPV treatment.
- **Methods:**
 - Designed wearable devices integrating multi-wavelength PPG sensors, skin temperature sensors, and IMU sensors
 - Developed dual-sensor configuration with reference monitoring for comparative tissue perfusion analysis
 - Created algorithms to detect ischemia/congestion from PPG signal components and guide head rotations for BPPV treatment
- **Results:**
 - Validated IMU-CRP system in clinical study (N=19) demonstrating improved BPPV treatment accuracy
 - Successfully implemented real-time monitoring with predictive analytics for flap procedures
 - [Published findings in Scientific Reports \(2023, SCIE Q1\)](#)
- **Skills developed:** Biomedical sensor integration, hardware design, Bluetooth communication, medical device certification protocols, IMU sensor signal processing, real-time monitoring systems, clinical validation methodologies



INDUSTRY EXPERIENCE

MedInTech:

- Designed and developed a real-time GPU monitoring web server
- MedInTech Gateway software for endoscopic video capture
- Designed and developed MD-SC-300 AI-powered cancer classification system

IDETECA S.R.L. :

- Evaluated medical device software/hardware components using specialized sensor technology
- Designed electrical/mechanical safety testing protocols compliant with IEC 60601 standards
- Verified sensor output signal stability/accuracy using oscilloscopes, data loggers, spectrum analyzers

IMEXCOMED S.R.L. :

- Performed preventive/corrective maintenance for endoscopy and surgical equipment
- Conducted component replacement, circuit testing, and on-site troubleshooting
- Provided installation services and advanced software support in hospital environments

Thank you

Cecilia Andrea Callejas Pastor
세실리아 칼레하스

PhD. In Biomedical Engineering

✉ cecilia.callejas.pastor@gmail.com

☎ +82 10-4302-9697